Adaptive Retrieval-Augmented Generation for Domain-Specific Document Understanding

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Abstract
Retrieval-Augmented Generation (RAG) has demonstrated remarkable performance in open-domain question

Keywords: Retrieval-Augmented Generation, Document Understanding, Domain Adaptation, Natural Languag

1. Introduction
The proliferation of domain-specific documents in enterprise environments has created an urgent need

Retrieval-Augmented Generation addresses hallucination by grounding LLM outputs in retrieved evidenc

In this paper, we present Adaptive-RAG, which addresses these limitations through three key innovati

2. Related Work
Prior work on RAG systems includes DPR (Karpukhin et al., 2020), REALM (Guu et al., 2020), and Atlas

3. Methodology
3.1 Domain Classification
We train a lightweight DistilBERT classifier on 50,000 annotated document headers to classify incomi

3.2 Adaptive Chunking
Each domain employs optimized chunking strategies:
- Legal: Clause-level chunking with section hierarchy preservation
- Medical: Entity-aware chunking maintaining diagnostic context windows
- Financial: Table-aware chunking with numerical precision preservation
- Technical: Code-block-aware chunking with API reference linking
- General: Semantic paragraph-level chunking

3.3 Retrieval Pipeline
We implement a hybrid retrieval approach combining dense retrieval (domain-fine-tuned Sentence-BERT)
Score = · × Dense(q,d) + (1-·) × BM25(q,d) + · × CrossEncoder(q,d)
where · and · are domain-specific hyperparameters.

3.4 Adaptive Prompt Compilation
Based on query complexity (measured by entity count and relation depth), the system selects from thr

4. Datasets
We evaluate on three benchmarks:
- ContractQA: 5,000 question-answer pairs from legal contracts
- MedDocQA: 3,200 clinical document understanding tasks
- FinanceExtract: 4,500 financial report extraction tasks

5. Results

Table 1: Main Results (F1 Score)
Method	ContractQA	MedDocQA	FinanceExtract	Average
Vanilla RAG	67.3	59.1	71.2	65.9
DPR + GPT-4	72.8	64.5	75.3	70.9

BM25 + GPT-4	69.1	61.8	73.7	68.2
Hybrid RAG	74.6	67.2	77.1	73.0
Adaptive-RAG	82.4	76.8	85.9	81.7

Adaptive-RAG achieves the highest F1 scores across all three benchmarks, with particularly strong im

Table 2: Faithfulness and Latency

Method	Faithfulness	Latency (ms)
Vanilla RAG	0.71	245
Adaptive-RAG	0.94	312

6. Key Findings

1. Domain-specific chunking is the single most impactful component, contributing 8.2% of the total F
2. Adaptive prompt compilation provides an additional 4.1% improvement, particularly for complex mul
3. The hybrid retrieval approach (dense + sparse + re-ranker) outperforms any single retrieval metho
4. Cross-domain transfer is possible: training on two domains and fine-tuning on a third achieves 95

7. Limitations and Future Work

Our approach currently supports five predefined domains. Extending to zero-shot domain adaptation re

8. Conclusion

We present Adaptive-RAG, a unified framework for domain-specific document understanding that dynamic

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